Hongming Xu

muzhihai@sjtu.edu.cn ◇ xuhm3410@gmail.com
GitHub ◇ Webpage

Research Interests

My research interests mainly lie in **coding foundation models, long-horizon agents, and self-improving AI**. I am now focusing on **agent memory, parametric memory, coding-model post-training, and recursive self-improvement**, especially on how models and agents can accumulate, update, and reuse experience over long-term interaction. I also have a background in **database systems and data management**, which led me to study memory from a data-management perspective and later move toward **agentic coding, long-horizon memory, and model self-improvement**.

Education

Shanghai Jiao Tong University — Shanghai, China
Undergraduate in School of Computer Science — Sept. 2024 – Present
Member of the Zhiyuan Honors Program

Selected Publications & Manuscripts

MemTrace: State-Consistent Memory for Long-Horizon Coding Agents

Under review at ICLR 2027 | **First author** **[arXiv]** **[Repository]** — Aug. 2026 – Present

Developed a state-consistent memory system that combines structured execution traces with repository grounding to validate historical evidence before reuse. Evaluated across three software-engineering benchmarks and two coding-agent harnesses, consistently improving the primary metric over native memory strategies.

Dial: A Knowledge-Grounded Dialect-Specific NL2SQL System

PVLDB 2026 | **Co-first author (listed second)** **[arXiv]** **[Code]** — Aug. 2025 – Mar. 2026

Contributed to natural-language preprocessing, dialect-aware logical planning, and knowledge retrieval for multi-dialect NL2SQL. Helped design and initialize **HINT-KB**, prepare datasets, and develop iterative feedback and semantic correction; participated in failure analysis and rebuttal revisions.

Are We Ready for An Agent-Native Memory System?

Under review at VLDB **[arXiv]** **[Code]** — Mar. 2026 – June 2026

Contributed to a systematic study of agent memory across representation and storage, extraction, retrieval/routing, and maintenance. Evaluated task effectiveness, retrieval fidelity, dynamic-update robustness, long-term stability, and runtime cost across diverse workloads.

Tensor-Network Physical Pages for Quantum-Ready Data Management

QCDKM 2026 — May 2026 – June 2026

Contributed to **TN-Page**, a tensor-network physical-page representation for approximate data management and quantum backends. Studied workload-aware bucketization, tensorization, factorization, and backend-aware encoding, with evaluation of fidelity, compression, qubit usage, and gate depth.

Research Experience

MemTensor

Research Intern — Mentor: Juncheng Zhang (Tsinghua University) — Aug. 2026 – Present

- Contribute to coding-oriented post-training of a **284B-parameter DeepSeek-V4-Flash-0731** foundation model. Design and synthesize software-engineering tasks, curate and filter training data and execution trajectories, and evaluate post-trained checkpoints on coding and repository-level tasks. Analyze capability gains, failure modes, and remaining skill gaps to guide subsequent data selection, task construction, and training iterations.
- Explore memory mechanisms for long-horizon coding agents, focusing on persistent execution history, repository-state alignment, and reliable reuse of historical evidence; led the first-author **MemTrace** project on state-consistent memory for repository-scale coding tasks.

Shanghai Jiao Tong University — Theseus Lab

Undergraduate Researcher — Supervisor: Prof. Xuanhe Zhou — Aug. 2025 – June 2026

- Worked on **Dial**, a knowledge-grounded dialect-specific NL2SQL system. Contributed to dialect-aware logical planning and the construction of **HINT-KB**, including knowledge organization, initialization, retrieval components, semantic correction, and failure analysis for heterogeneous SQL dialects.
- Extended this data-management perspective toward **agent memory and memory management** through *Are We Ready for An Agent-Native Memory System?*. Studied memory representation and storage, extraction, retrieval, maintenance, and systematic evaluation, forming the basis for my later work on long-horizon coding-agent memory.

Honors

Zhiyuan Honors Scholarship

Personal Interests

Badminton and fitness.